

Double Patterning and Hyper-Numerical Aperture Immersion Lithography

In attempt to continue the advancement of fabrication in accordance with Moore's Law, double patterning lithography (DPL) has been introduced. Double patterning lithography does not introduce any new technology and is used because the industry requires it, rather than actually desires to use it [1]. Since no new technology is introduced, it is viewed as a short-term solution. Double patterning lithography is useful in that it will allow fabrication of the 22nm node and beyond [1]. There are several double patterning lithography techniques that have been developed. These techniques are litho-etch litho-etch (LELE), litho-freeze litho-etch (LFLE), and self-aligned double patterning (SADP), each with its own advantages and disadvantages. The most commonly used techniques are LELE and LFLE.

The first double patterning technique, litho-etch litho-etch (LELE), is the most straightforward. It consists of an initial lithography step followed by an etching step and then the same process is repeated to get the final pattern, see figure 1. The negative aspects of this technique are that two masks must be used to create the final pattern. This directly increases the time and cost it takes to fabricate using double patterning because the process steps will be effectively doubled [2]. Another struggle with this technique is small tolerance of the pattern overlay [2]. Because of the small-scale patterns used with DPL, a slight misalignment of the second pattern will cause a notable defect in the final pattern.

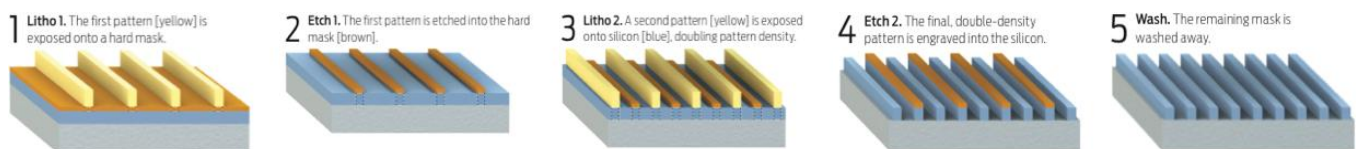


Figure 1. LELE Process [3].

The second technique, litho-freeze litho-etch (LFLE), was introduced to eliminate the increased cost due to additional process steps of the LELE technique. In LFLE there are still two lithography steps, but only one etching step. This can be done through the 'freezing' of the first pattern [2]. The freezing step is implemented in such a way that a chemical treatment is able to alter the initial pattern so that it is unaffected by the exposure and development of the second pattern [2]. Following the development of the second mask, a single etching step takes place. As a result, LFLE is cheaper to perform than LELE, but still faces issues with small pattern alignment tolerances.

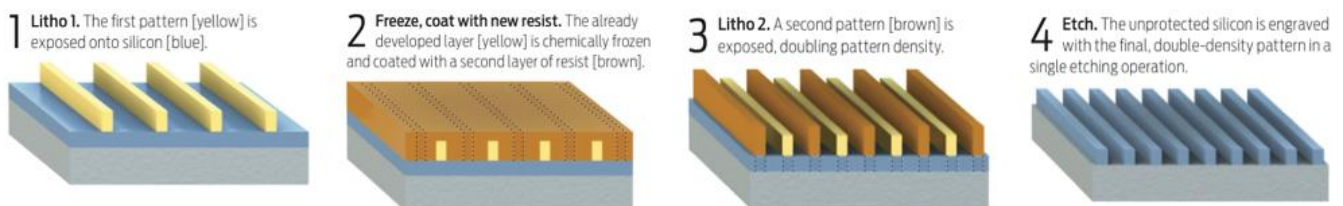


Figure 2. LFLE Process [3].

The technique of self-aligned double patterning (SADP) eliminates the complication of pattern alignment seen in LELE and LFLE [2]. As shown in figure 3, SADP uses an initial photoresist pattern to

determine the spacing of the final pattern, which will be etched. Once the initial photoresist is exposed and developed, a masking material is deposited and etched, forming sidewall spacers. After removing the initial photoresist, the pattern is etched using the remaining sidewall spacers. This technique is more accurate than the first two because it is easier to control pattern dimensions than pattern placement and the use of the initial photoresist pattern fixes the placement of the final pattern leaving only the dimension of the sidewall to maintain. Although it reduces the overlay problems associated with LELE and LFLE, SADP is only suited for uniform patterns for applications such as memory, not logic which is often irregular [2]. Additionally, the cost of this technique is increased with the added process steps used to deposit, etch, and remove the sidewall spacers.



Figure 3. SADP Process [3].

Currently, there is a clear separation in applications of double patterning lithography in the integrated circuit industry. The separation is made between the fabrication of memory and logic. Memory fabrication most commonly uses self-aligned double patterning (SADP) because of its repeating pattern. While logic fabrication is implemented through litho-etch litho-etch (LELE) and litho-freeze litho-etch (LFLE) because of its tendency to have non-uniform patterns. This separation in application of DPL is the reason memory manufacturers such as Hynix, Micron, Renesas, and Samsung utilize SADP. Whereas the IBM partnership companies, Intel, Sony, TI, Toshiba, and TSMC, who manufacture logic devices use LELE and LFLE [2].